

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/277597419>

Use of Information Networks and Resource Sharing in Digital Age

Technical Report · October 2014

DOI: 10.13140/RG.2.1.2981.6489

CITATIONS

0

READS

119

1 author:



Mohammed Usman

Institute of Human Virology

6 PUBLICATIONS 0 CITATIONS

SEE PROFILE

Some of the authors of this publication are also working on these related projects:



Information awareness as a tool for preventing HIV/AIDS among the University students of Umaru Musa Yar'dua Katsina, Nigeria. [View project](#)

“USE OF INFORMATION NETWORKS AND RESOURCE SHARING IN DIGITAL AGE”

INTRODUCTION

Libraries have traditionally co-operated within information networks, usually of an informal character, in order to provide the most cost-effective services to their clients. This cooperation has been based on arrangements for the exchange of publications, inter-library lending, supply of photocopies etc. Whereas the resources of information are growing and diversifying at an ever increasing speed, it is unfortunately rare that the financial resources needed to acquire or access them are growing at the same rate. Maintaining an information resource in support of research and management activities has always been costly. The recent adoption of information as a commodity in a so-called “information society” will probably not render it less expensive.

The need for even greater co-operation between libraries, information networks) for providing access to and sharing the available resources, is essential in regions where institutions and organization do not have a strong information base or adequate funding to obtain the wealth of information available.

INFORMATION

Information (shortened as **infor.** or **info.**) is that which informs, i.e. an answer to a question, as well as that from which knowledge and data can be derived (as data represents values attributed to parameters, and knowledge signifies understanding of real things or abstract concepts). As it regards data, the information's existence is not necessarily coupled to an observer (it exists beyond an event horizon, for example), while in the case of knowledge, information requires a cognitive observer.

At its most fundamental, information is any propagation of cause and effect within a system. Information is conveyed either as the content of a message or through direct or indirect observation of something. That which is perceived can be construed as a message in its own right, and in that sense, information is always conveyed as the content of a message.

Information can be encoded into various forms for transmission and interpretation (for example, information may be encoded into signs, and transmitted via signals). It can also be encrypted for safe storage and communication.

INFORMATION NETWORKS

An information networks is a set of two or more computer connected together to share information and resources like printer and hard disk etc. it is also called Information network. *Martin & Thomas (2011)*

The information network uses two important technologies. These are computing and telecommunications. Telecommunication is a process of transferring information over a distance through radio waves and optical signals etc. the computer in a network are connected with one another through cables, satellite or phone lines.

According to *Peterson LL, Davie BS. (2012)* Information networks or data networks is a telecommunications network that allows computers to exchange data. Information networks, networked computing devices pass data to each other along data connections (network links). Data is transferred in the form of packets. The connections between nodes are established using either cable media or wireless media. The best-known Information networks are intranet and Internet.

Information networks differ in the transmission media used to carry their signals, the communications protocols to organize network traffic, the

network's size, topology and organizational intent. In most cases, communications protocols are layered on (i.e. work using) other more specific or more general communications protocols, except for the *physical layer* that directly deals with the transmission media.

Information networks support applications such as access to the World Wide Web, shared use of application and storage servers, printers, and fax machines, and use of email and instant messaging applications.

COMPUTER NETWORKS

A computer network, or simply a network, is a collection of computers and other hardware components interconnected by communication channels that allow sharing of resources and information. Today, computer networks are the core of modern communication. All modern aspects of the public switched telephone network (PSTN) are computer-controlled. Telephony increasingly runs over the Internet Protocol, although not necessarily the public Internet. The scope of communication has increased significantly in the past decade. This boom in communications would not have been possible without the progressively advancing computer network. Computer

networks, and the technologies that make communication between networked computers possible, continue to drive computer hardware, software, and peripherals industries. The expansion of related industries is mirrored by growth in the numbers and types of people using networks, from the researcher to the home user. *Peterson LL, Davie BS. (2011).*

INFORMATION NETWORKS AND RESOURCE SHARING IN DIGITAL AGE

As digital information supplements and even replaces print material, librarians must adapt resource sharing policies, practices, and procedures to the technological, economic, and legal realities of the digital age. Although interlibrary loan (ILL) specialists provide an invaluable service, this does not mean that everything they do is right, beneficial or good for their patrons, or for society as a whole. As with any human endeavor, they should neither assume that their work is immune to ethical considerations, nor should they take ethical shortcuts. Otherwise, even the most well-meaning librarian risks compromising their values, interests, and mission, as well as harming other stakeholders in the information world, including their own constituents.

Traditional library and information ethics and librarian values stem from established ethical frameworks, which adds to their power. Established

principles of professional librarianship acknowledge the rights of information producers, the importance of the free exchange of knowledge for social progress, and the responsibility for preserving knowledge for posterity. Despite cross-cultural and political differences, (Vaagan, 2002) IFLA's Committee on Free Access to Information and Freedom of Expression (FAIFE) commission also expresses this in an international context. As for the values of resource sharing professionals, reciprocity and cooperation are both fundamental.

Many of the practical challenges facing librarians, however, are new. As more information is created, put online, and made accessible instantaneously, information sharing is technically easier, yet digital technology is also positioning publishers to gain more control over the creation, dissemination and preservation of information. (Smith, 2011; Geffert, 2011; Gleason, 1988)

As for resource sharing, adhering to differing copyright and licensing laws around the world is challenging, as is operating under increasingly significant financial constraints. (Hadro, 2010) Librarians want to support

international borrowing (and lending), and people are requesting it, but if libraries and patrons cannot afford it, or are not permitted to share with libraries around the world, then they may have to do without information that they need. (*Smith*, 2011) Resource sharing staff must also minimize the negative environmental impact of digital resource sharing procedures, such as when publishers require the printing of online material, rather than allowing it to be transmitted electronically. Most important, though, is the need to protect and extend the legal concept of the fair use of digital information for educational and cultural advancement, rather than accepting license terms that unduly restrict interlibrary loans. (*Poynder*, 2011; *Massie*, 2010)

It may come as a great surprise that the values guiding librarians are not universally embraced by publishers, politicians, authors, the public, or even all librarians, but other stakeholders do have other interests. This is why, in order to convince others to support their work, librarians should reference the moral arguments that philosophers have devised as they develop new ways to practically and ethically navigate the digital landscape.

CHARGING FEES FOR INFOR. NETWORKS AND RESOURCE SHARING

Online catalogs and websites now make information discovery much easier, but the full text of much of it is not available for free through libraries. Publishers are now charging libraries more money to access information online, even when they have already purchased it in print. (Poynder, 2011; Wright, 1997) Digital information can also be more expensive when it is only available through a publisher, because no library has bought or licensed it, or because they have but are not permitted to use it to fill ILL requests. ILL services can also be increasingly costly when using commercial document supply companies that charge a premium for speed. More libraries are also now charging fees to other libraries in order to recoup some of the costs of lending.

Any user who claims a "right" to free information or services is arguing for an idealistic extreme. However, librarians are committed to access and serving those who cannot afford to pay, as well as those who can. In effect, such access is often subsidized through grants, donations, a progressive income tax, or tuition charges adjusted for need based financial aid. Governments could also do more to fund libraries, if they deem them a common public good. Alternatively, or in addition, lawmakers could legislate more liberal fair use doctrines and educational exceptions to

copyright laws so that libraries could share more information. Or, publishers could find cheaper pricing options for libraries to access online e-articles and e-books on a small per use basis, rather than the prices they now charge, which are often as high as the price to buy an entire print book, or the rigid and expensive big packages that they offer to libraries.

Of course, libraries can join consortia that agree to freely lend to members. (Although lending requests may then increase, this can be mitigated by other efficiencies.) The use of automated technology is another way to decrease the cost per transaction, although there are startup costs for purchasing management software, such as ILLiad, as well as technical support costs. Some products, such as the Odyssey Standalone document transmission software are even free. Unless costs are controlled, when it is more costly for libraries to provide services, librarians may have to pass along fees to patrons. The danger, though, is that in many cases this means that patrons will choose to make do without the information, limiting the advance of knowledge, which is a common good, as well as a right, with a great deal of utility.

CONCLUSION

Many academics, institutions, national and international libraries have or are developing open digital libraries for information networks and resource sharing , therefore librarians should continue to create, support, and market projects that provide digital information to the public for no direct cost.

REFERENCES

- ARL (1995), "Fair use in the electronic age: serving the public interest", Associates. ISBN 1-56592-660-9.
- Aufderheide, P. and Jaszi, P. (2011), *Reclaiming Fair Use: How to Put Balance Back in Copyright*, The University of Chicago Press, Chicago and London, pp. 148-151.
- Beth Posner. (2012) "*The Ethics of Library Resource Sharing in the Digital Age*" The CUNY Graduate Center
- Bentham, J. (2011), *An Introduction to the Principles of Morals and interlibrary loans*", *Interlending and Document Supply*, Vol. 25 No. 2, pp. 51-56.
- Martin, Thomas. "Design Principles for DSL-Based Access Solutions".
- Peterson LL, Davie BS. (2011). *Computer Networks: A Systems Approach*
- Poynder, R. (2011), "The big deal: not price but cost", *Information Today*, Retrieved 18 June 2011.
- Spurgeon, Charles E. (2000). *Ethernet The Definitive Guide*. O'Reilly & University Press, Cambridge.
- Vaagan, R. W. with an introduction by Byrne, A. (2002), *The Ethics* Vol. 28 No. 8, September,
- Wattles, J. (1996), *The Golden Rule*, Oxford University Press, Oxford.
- Williams, B. (2000; 1972), *Morality: An Introduction to Ethics*, Cambridge
- Wood, Jessica (2010). "The Darknet: A Digital Copyright Revolution".
- Wright, C. (1997), "From barter to business: a moral history of charging for

information resource